Lab 1 - PECARN TBI Data, STAT 215A, Spring 2025

September 20, 2025

1 Introduction

Traumatic brain injury (TBI) is one of the leading causes of death and disability in children. In the United States, head trauma accounts for thousands of deaths, tens of thousands of hospital admissions, and more than half a million emergency department visits each year [1]. When a child presents after head trauma, clinicians must weigh two competing priorities: on one hand, the need to rapidly detect clinically-important traumatic brain injuries (ciTBI), and on the other hand, the desire to avoid unnecessary computed tomography (CT) scans. While CT is the reference standard for diagnosing TBI, it exposes children to ionizing radiation, which carries a measurable risk of malignancy. This trade-off makes it essential to identify children at very low risk of ciTBI, for whom CT can safely be avoided.

The PECARN dataset, collected across 25 pediatric emergency departments and analyzed by Kuppermann et al. (2009) [1], addresses this problem directly by developing and validating age-specific clinical decision rules for ciTBI. These rules are simple and interpretable, and have been widely adopted in practice, but the underlying dataset contains many more variables than were used in the original publication. Exploring this dataset allows to better understand both the clinical patterns of pediatric head trauma and the challenges of working with real-world medical data, such as missingness, inconsistent coding, and the need for interpretability in high-stakes decision making.

From a data science perspective, this analysis connects closely to the PCS (predictability, computability, stability) framework from Bin Yu's *Veridical Data Science* [2]. PCS emphasizes that reliable conclusions require careful data cleaning, close connection to domain knowledge, and stability checks to ensure that findings are not artifacts of arbitrary choices. This is particularly relevant here, since any data-driven rule for ciTBI must balance sensitivity (avoiding missed injuries) with practicality and clarity for clinicians.

The purpose of this exploratory data analysis (EDA) is therefore: (1) to clean and document the PECARN dataset, (2) to explore key variables and their relationships with ciTBI outcomes, and (3) to frame these findings in the clinical context of CT decision making. The remainder of this report is structured as follows: Chapter 2 introduces the dataset, explains how it was collected, and describes the data cleaning process (2.2), followed by a first look at the variables (2.3). Chapter 3 presents two main exploratory findings, supported by publication-quality graphics, and includes a reality check (3.3) and stability check (3.4). Chapter 4 implements and compares two models for predicting ciTBI, and examines their stability under perturbations from Chapter 3.4. Chapter 5 concludes with a discussion of the analysis through the PCS lens, reflecting on the implications for both data science and clinical practice.

2 Data

The dataset analyzed in this report comes from the Pediatric Emergency Care Applied Research Network (PECARN), as published in [1]. The full cohort included more than 43,000 children younger than 18 years who presented to one of 25 North American emergency departments within 24 hours of head trauma, across the full range of Glasgow Coma Scale (GCS) scores (3–15). However, for the purposes of developing and validating prediction rules, the study focused on the subgroup of 42,412 children with GCS scores of 14 or 15, representing cases of minor head trauma. Children with GCS scores of 3–13 were analyzed separately, since their substantially higher risk of traumatic brain injury makes CT scanning the clear clinical standard and does not pose the same decision dilemma.

The dataset includes a wide range of variables describing demographics, mechanism of injury, clinical symptoms and signs, and outcomes. The dataset also records whether a CT scan was performed, whether traumatic brain injury was observed on imaging, and whether the case qualified as a ciTBI according to PECARN's definition.

The central question is how to identify children at very low risk of ciTBI so that unnecessary CT scans can be avoided. The richness of the PECARN dataset allows us to investigate which patient characteristics are most strongly associated with ciTBI, how well simple decision rules perform, and where potential improvements or refinements might be possible. The goal is to balance the benefit of detecting serious injuries against the harm of exposing children to radiation.

2.1 Data Collection

The data was collected as part of the large prospective cohort study by the Pediatric Emergency Care Applied Research Network (PECARN) [1]. Children younger than 18 years presenting with head trauma within 24 hours of injury were enrolled across 25 North American emergency departments between 2004 and 2006. Trained clinicians recorded patient histories, clinical symptoms, and wrote down their findings on a standardized case report forms before knowing the CT scan results. Follow-up information was obtained by reviewing hospital records and, in discharged cases, by conducting telephone follow-up with parents to identify any missed injuries.

The dataset contains variables spanning several domains:

- **Demographics:** age (in months and years), sex, and other baseline characteristics.
- **Mechanism of injury:** categorized into low, moderate, high, as well as into groups such as motor vehicle collision, fall, sports injury, assault, or other blunt trauma.
- **Clinical assessments:** Glasgow Coma Scale (GCS) score, mental status, neurological deficits, and presence of scalp hematoma.
- **Symptoms:** loss of consciousness (with recorded duration), vomiting (including frequency), headache, dizziness, and behavioral changes reported by parents.
- **Outcomes:** whether a CT was performed, TBI findings on CT, and classification as clinically-important TBI (ciTBI), defined by outcomes such as death, neurosurgery, intubation, or hospital admission.

The measurement of these variables naturally varies in precision and subjectivity. For example, GCS scores are standardized but may differ slightly depending on the examiner, while reports of symptoms like headache or vomiting rely on patient or parental reporting and are less objective. Injury mechanisms are recorded categorically but may vary in detail (e.g., type of fall or vehicle speed). Similarly, some physical signs such as scalp hematoma size are more difficult to measure consistently, leading to variability in the data. These differences highlight why careful cleaning and exploration of the dataset are necessary steps before drawing reliable conclusions.

2.2 Data Cleaning

Initial Variable Selection

Before cleaning the entire dataset, a smaller set of predictors was chosen as a starting point. The intention was to first gain familiarity with variables that appeared clinically important, were relatively straightforward to interpret, and connected directly to the PECARN decision rules [1]. This reduced set provided a manageable entry point into the data. The guiding motivation was not to prematurely discard information, but to focus on variables most likely to be associated with clinically-important traumatic brain injury (ciTBI) and to return to more detailed variables if the simplified versions proved insufficient.

Motivation for variable reduction. To improve robustness in the analysis, the variables were first grouped into four categories: participant data, hard fact symptoms, reported categories, and outcomes/decisions, with the motivation that clinician-observed "hard facts" (e.g., GCS, palpable skull fracture) are generally less subjective than self-reported symptoms such as headache or dizziness, which may vary considerably across patients and observers. In the following the initial variable selection will be discussed.

For participant data, `AgeInMonth` and `AgeTwoPlus` were prioritized because age stratifies PECARN pathways and risk; `Gender` and `Race` were retained as basic demographics that may reflect contextual differences in presentation and care, even though they are not core PECARN predictors.

Among hard-fact symptoms, variables were chosen, that are clinical standards, easy to measure and report, and seemed important in the context. `GCSTotal` was emphasized first because it is a standardized neurological scale (3–15) and central to triage; `AMS` (altered mental status) was included as a key clinical flag, recognizing that it can be variably assessed but is clinically consequential, and it was treated conservatively in combination with `GCSTotal`. `SFxPalp` (palpable/basilar skull fracture signs) was kept as a high–specificity marker of serious injury. External trauma indicators were retained: `Hema` (any scalp hematoma) and `HemaSize` (size category), acknowledging that size indicates the severity of the accident; similarly, `Clav` is used to localize trauma above the clavicles. `NeuroD` (any focal neurological deficit) was included despite low prevalence because its presence is strongly predictive. `Seiz` (post-injury seizure) was included as another high-concern sign.

For reported categories, especially variables that are easy to answer (Yes/No) were taken into account with the caveat that some may be subjective. `ActNorm` (acting normal) was used as a pragmatic proxy for behavior change, especially informative in younger children; `Amnesia_verb` captured post-traumatic amnesia with the caveat of self/parent report; `Dizzy` was kept as a subjective symptom that can still carry signal in aggregate. Headache variables were included because they are both common and part of decision rules: `HA_verb` (any headache), `HASeverity` (severity level), and `HAStart` (onset timing), with recognition that severity and timing are more subjective. Vomiting was represented by `Vomit` (any) and `VomitNbr` (count category), since greater frequency can raise concern but reporting can vary. Mechanism was initially simplified by using `High_impact_InjSev` rather than enumerating all detailed mechanism codes; this clinician-coded grouping into {Low, Moderate, High} provides an interpretable measure for trauma energy without requiring expert knowledge, but its subjectivity was acknowledged and the plan set to revisit finer mechanism detail if analyses suggest misclassification or instability. Finally, consciousness events were captured with `LOCSeparate` (No/Yes/Suspected) and `LocLen` (duration categories) to reflect both occurrence and rough severity of loss of consciousness, noting that brief or uncertain episodes can be reported inconsistently.

Outcomes and actions were placed last: `CTDone` was kept to study imaging decisions (potential over-/under-use) and `PosIntFinal` (ciTBI composite) was retained as the primary endpoint. This staged reduction favors standardized, high-signal variables early (e.g., `GCSTotal`, `SFxPalp`, `NeuroD`) while preserving subjective but clinically relevant reports (e.g., `HA_verb`, `Vomit`, `Amnesia_verb`) for later checks. If simplified choices such as `High_impact_InjSev` prove unstable or misleading, a return to more granular encodings is planned.

A tabular overview of the reduced set of variables is summarized in Table 1. Note that the variable names are different, as some were renamed, which is discussed in Step 3.

This reduction allowed for an initial exploration of clinically relevant signals without being overwhelmed by the full dimensionality of the dataset. The strategy follows the PCS principle of beginning with interpretable, domain-grounded variables and iterating as needed [2].

The raw PECARN dataset was examined systematically before cleaning, following the PCS principle of "examine and create action items" [2]. Several issues were identified: sentinel codes used in place of missing values, inconsistent naming of variables across sources, overlapping categories, and heterogeneous data types. The goal of cleaning was to make these inconsistencies explicit, transform variables into interpretable and reproducible forms, and preserve missingness where appropriate.

Step 1: Check completeness and identify missingness. Several variables use sentinel codes (e.g. 91, 92) to represent "unknown" or "not applicable." Examples include `Amnesia_verb`, `HA_verb`, `HASeverity`, `VomitNbr`, and `HemaSize`. These codes were systematically recoded to `NaN` to ensure that analyses would not misinterpret them as valid numerical values. Here the judgment call was made to treat "unknown" or "not applicable" the same as a fully missing variable, since it is reasonable to assume that missing values were not written down, because they were not applicable or not important in that case.

Step 2: Verify ranges and data validity. Variables with natural ranges were validated. For **Age**, values outside 0–216 months (18 years) were checked. All value were in the valid range. Otherwise they'd have been set to `NaN`. For **GCS**, valid scores are restricted to 3–15; values outside this range were discarded. These checks ensure that implausible entries (e.g. negative ages or GCS outside the clinical definition) were excluded. Categorical variables such as **HeadacheSeverity**, **HematomaSize**, and **VomitNbr** were mapped only within their defined categories; any unexpected codes and missing values were set to `NaN`.

Step 3: Rename messy labels and categories. Numeric encodings were mapped to meaningful clinical labels. For example, **Gender** (1=Male, 2=Female) was mapped to string labels with "Unknown" as fallback. **Injury severity** (`High_impact_InjSev`) was mapped into {Low, Moderate, High}. To facilitate modeling, a binary indicator (`InjurySeverityHigh`) was derived to distinguish high-impact mechanisms from others. This decision may be revisited, as someone could have a very soft motor vehicle collision (classified as `High`) that is less strong than running with a high speed into an object (classified as `Low`). In addition, binary indicators such as **HematomaSizeML** (Medium or Large vs. Small) were created to reduce dimensionality of categorial variables.

Step 4: Resolve variable redundancy and synonyms. No true redundancies or duplicate synonyms were found in the dataset. However, some variables were constructed to reduce dimensionality of more detailed information, such as `High_impact_InjSev`, which summarizes the full injury mechanism codes into three levels (Low, Moderate, High). In addition, several variables were renamed to make them easier to use and interpret consistently. The main renamings are summarized in Table 1.

Original	**New name**	**Original**	**New name**
`AgeInMonth`	Age	`HASeverity`	HeadacheSeverity
`AgeTwoPlus`	AgeTwoPlus	`HAStart`	HeadacheStart
`Gender`	Gender	`Vomit`	Vomit
`Race`	Race	`VomitNbr`	VomitNbr
`High_impact_InjSev`	InjurySeverity	`Dizzy`	Dizzy
`Amnesia_verb`	Amnesia	`GCSTotal`	GCS
`LOCSeparate`	LOC	`AMS`	AlteredMentalStatus
`LocLen`	LOCgt5s	`SFxPalp`	SkullFracture
`ActNorm`	ActAbnormal	`Hema`	Hematoma
`Seiz`	Seizure	`HemaSize`	HematomaSize
`HA_verb`	Headache	`Clav`	Clavicles
`NeuroD`	NeurologicalDeficit	`ClavFace,...`	ClaviclesTyp
`NeuroDMotor,...`	NeurologicalDeficitTyp	`CTDone`	CTDone
`PosIntFinal`	CiTBI		

Table 1: Renamed variables for ease of use and interpretability.

Step 5: Derived clinical indicators. Composite predictors were derived from raw fields to align with the PECARN rules.

- **LOC** from LOCSeparate with levels {No, Yes, Suspected}, plus a binary indicator LOCbinary (Yes/Suspected vs. No).
- **LOCgt5s** from LocLen, where loss of consciousness longer than five seconds was coded as True.
- **AlteredMentalStatus (AMS)** from AMS, with all patients having GCS $<$ 15 automatically set to True to reflect clinical practice.

Step 6: Standardize outcomes and Incomplete data. The primary outcome, **ciTBI** (PosIntFinal), was standardized as boolean. **CTDone** was also standardized. To ensure interpretability and comparability, cases with missing **Age**, **GCS**, **AMS**, **SkullFracture**, or **ciTBI** were dropped. This resulted in a total of 427 removed rows, i.e. a total cleaned data set size of 42972. This decision is reasonable and align with the number of ca. 500 dropped rows from [1].

A full list of variables in the cleaned data frame are shown in Table 3.

Variable	Type/Values	Unit/Scale	Short description	Meaning (clinical context)
Participant data				
Age	Int64 (nullable); 0–216	months	Child's age in months (values outside range → NaN)	Key risk stratifier; PECARN rules differ by age group
AgeTwoPlus	boolean	indicator	$\geq$2 years (True) vs <2 (False)	Encodes the PECARN age split
Gender	categorical	Male/Female/Unknown	Patient's sex as reported	Demographic descriptor
Race	categorical	major groups	Self-reported race, mapped from codes	Demographic/context variable
Hard fact symptoms (objective clinical findings)				
AlteredMentalStatus	boolean	indicator	AMS flag (forced True if GCS<15)	Captures confusion, agitation, slow responses
Amnesia	boolean	indicator	Any post-traumatic amnesia	Suggests altered consciousness
Clavicles	boolean	indicator	Any trauma above clavicles	Proxy for external head impact
GCS	Int; 3–15	scale	Glasgow Coma Scale	Gold standard measure of consciousness
Hematoma	boolean	indicator	Any scalp hematoma	High risk esp. <2 years
HematomaSize	categorical	Small/Medium/Large	Hematoma size code	Larger → more suspicious
HematomaSizeML	boolean	indicator	Medium/Large vs Small	Thresholded size variable
NeurologicalDeficit	boolean	indicator	Any focal neurological deficit	High specificity red flag
Seizure	boolean	indicator	Post-injury seizure	Strong sign of serious injury
SkullFracture	boolean	indicator	Signs of skull fracture	Strong predictor of ciTBI
Vomit	boolean	indicator	Any vomiting	Symptom indicator
VomitNbr	categorical	Once/Twice/>2	Number of vomiting episodes	Higher counts → higher suspicion
Reported categories (subjective symptoms / caregiver reports)				
ActAbormal	boolean	indicator	Acting abnormally per caregiver	AMS proxy in preverbal children
Dizzy	boolean	indicator	Report of dizziness	Subjective symptom
Headache	boolean	indicator	Any headache reported	Symptom indicator
HeadacheSeverity	categorical	Mild/Moderate/Severe	Self/parent-reported severity	Severity proxy, subjective
HeadacheStart	categorical	ordered labels	Onset timing (before, within 1h, etc.)	Symptom timing, often uncertain
InjurySeverity	categorical	Low/Moderate/High	Injury mechanism severity, grouped by PECARN	Proxy for trauma energy; subject to coder interpretation
InjurySeverityHigh	boolean	indicator	High vs Low/Moderate	Simplified flag for severe mechanisms
LOC	categorical	No/Yes/Suspected	Loss of consciousness status	Symptom of possible intracranial injury
LOCbinary	boolean	indicator	Yes/Suspected vs No	Conservative simplification of LOC
LOCgt5s	boolean	indicator	LOC duration >5s	Longer LOC indicates higher risk
Outcomes / actions				
CTDone	boolean	indicator	CT obtained in ED	Action variable (clinical decision)
CiTBI	boolean	indicator	Composite outcome	Clinically-important TBI (death, neurosurgery, ICU, $\geq$2-night stay)

Table 3: Cleaned variable dictionary used for analysis. Short columns are fixed in width while the description and meaning columns expand to use the remaining space. The outcome (`CiTBI`) and the action variable (`CTDone`) are listed at the end because they are targets for modeling and evaluation.

2.3 Data Exploration

Demographics. In this section the dataset will be explored at a descriptive level in order to build intuition about its structure and content before moving to more detailed findings. The exploration begins with general participant data such as age and gender. After that, the focus shifts to boolean predictors (e.g., presence or absence of symptoms and clinical signs), followed by categorical and score-based predictors. The goal is to gain an overview of how the data are distributed and to identify initial patterns that might later be correlated with clinically-important traumatic brain injury (ciTBI).

The distribution of age and gender is shown in Figure 1. The overall sample consists of 62.3 % male and 37.7 % female patients, while no cases were coded as gender unknown. This imbalance toward male participants is consistent with clinical observations that boys more often present with head trauma. The age distribution further shows that a large proportion of cases occur in very young children, particularly in the first years of life. This is an indicator of age-specific decision rules, since risk factors and clinical presentations can differ between infants, toddlers, and older children. In addition to the overall age and gender distributions, the same plots were created separately for the ciTBI positive and ciTBI negative groups. These subgroup distributions closely mirror the total distribution, suggesting that the age and gender profiles of patients with ciTBI do not differ substantially from those without.

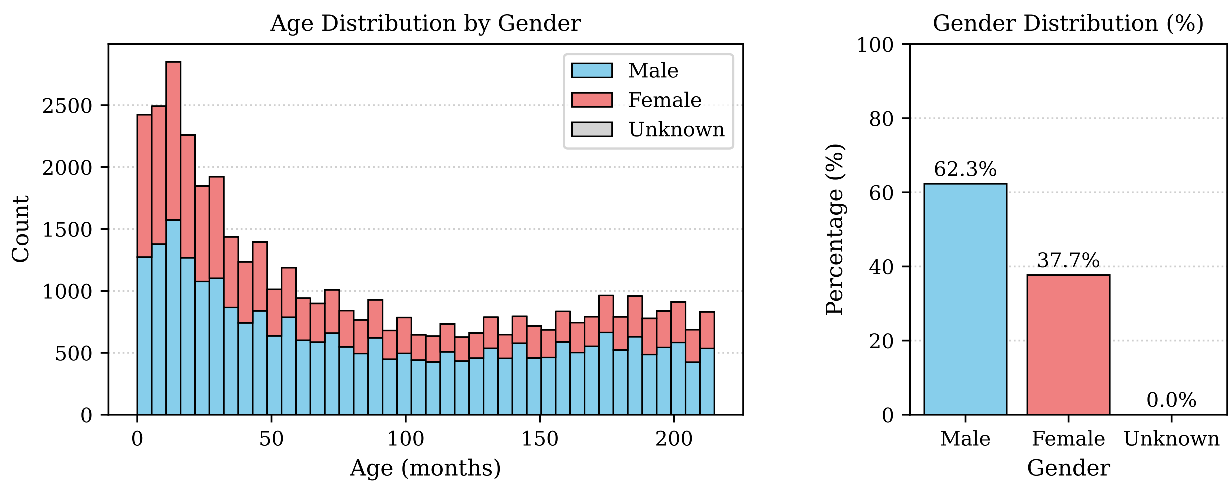


Figure 1: Age and gender distribution of the study population. The age histogram shows a concentration of cases in the first years of life, reflecting the high vulnerability of very young children to head trauma. The stacked bars indicate that both male and female patients are well represented across the age range, with a slight male predominance.

Boolean Variables. Figure 2 shows the distribution of boolean predictors stratified by ciTBI outcome, separately for children younger than two years and for those two years or older. For each predictor, the bars are split into "No ciTBI" on the left and "ciTBI" on the right, and the proportions of `False`, `True`, and missing values are shown within each outcome group.

Among the clinical signs, variables such as `AlteredMentalStatus`, `SkullFracture`, and `NeurologicalDeficit` are strongly associated with ciTBI across both age groups: the prevalence of "True" values is clearly higher in the ciTBI group. This is consistent with their established role in the PECARN decision rules. By contrast, variables like `Clavicles` or `HematomaSizeML` show weaker separation and may be less useful for predicting ciTBI in isolation.

Age-related differences are also visible. In older children, subjective symptoms such as `Headache` and `Amnesia` become more informative, as expected since preverbal children cannot reliably report these. Other reported symptoms like `Dizzy` or `ActAbnormal` show little distinction between ciTBI outcomes, suggesting limited predictive value. Gender-specific differences were investigated, but no significant trends were found. Therefore the following investigations won't distinguish between genders.

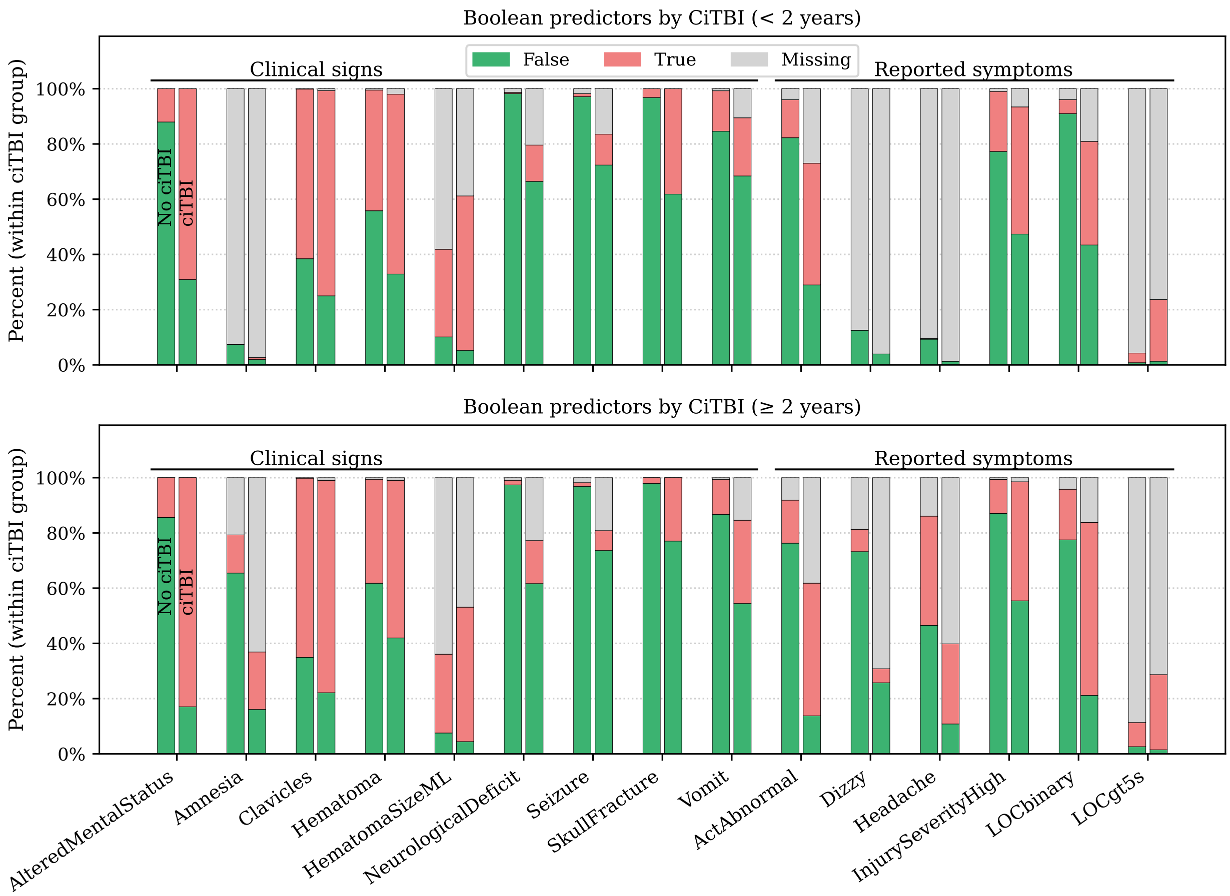


Figure 2: Boolean predictors by ciTBI outcome, stratified by age group. Each bar shows the percentage of `False`, `True`, and missing values within the "No ciTBI" and "ciTBI" groups. In older children, subjective symptoms such as headache and amnesia become more informative. Other predictors (e.g., dizziness, clavicle trauma) show little difference between outcome groups.

Overall, this exploratory plot highlights which boolean predictors align with established clinical risk factors and which may contribute little signal. It also shows the importance of age stratification: some predictors matter more in young, while others (e.g., headache, amnesia) become relevant only in older children.

False positive. To further investigate why CT scans were performed in children without ciTBI, predictor prevalences were compared between the unnecessary CT group (CT done, no ciTBI) and the necessary CT group (CT done, ciTBI) (Figure 3). The results show that symptoms such medium and large hematomas, headache, and clavicles were common among children who received CTs unnecessarily, whereas strong clinical predictors such as altered mental status, acting abnormal, and loss of consciousness were much more prevalent among children with ciTBI.

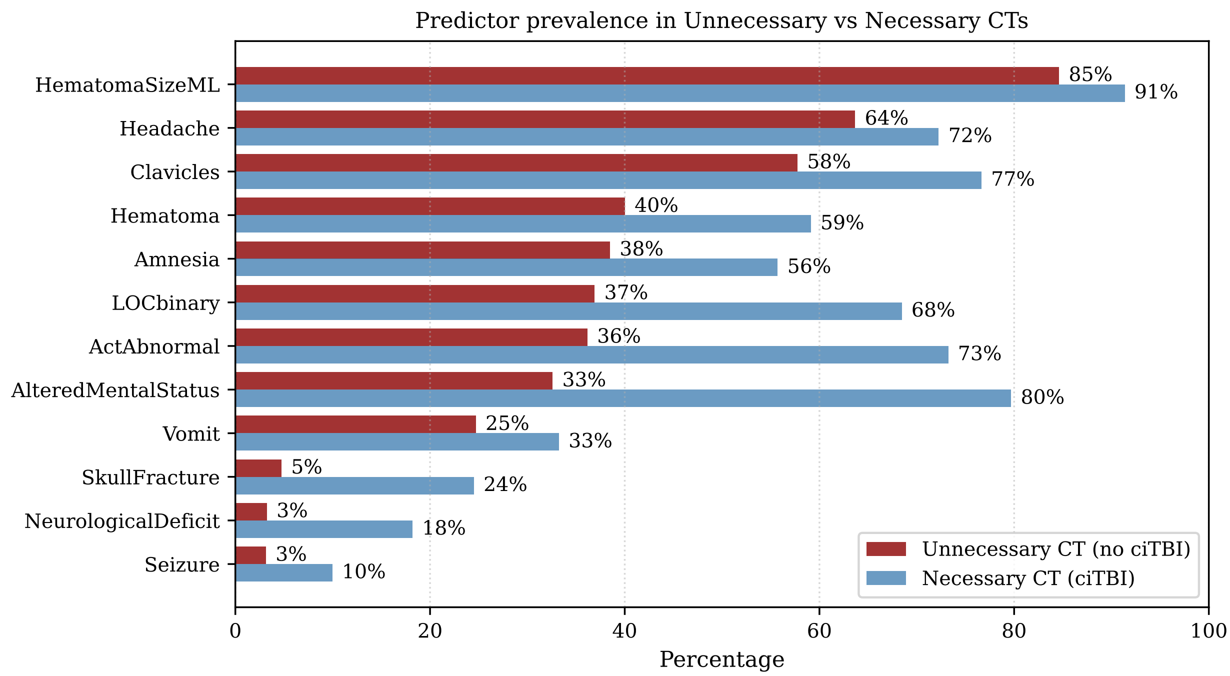


Figure 3: Prevalence of clinical predictors among children with CT but no ciTBI (unnecessary CTs, red) versus children with CT and ciTBI (necessary CTs, blue).

Numerical Predictors: Glasgow Coma Scale. Figure 4 displays the prevalence of ciTBI across different Glasgow Coma Scale (GCS) scores. While small differences between children younger than two years and those aged two years or older were observed in some predictors, the overall age-stratified patterns were not strong for GCS. Therefore, the results are presented for the entire cohort combined. The figure shows a clear gradient: ciTBI is extremely rare at GCS 15, but the prevalence rises steadily with decreasing GCS, reaching very high proportions for severely impaired patients. This confirms that GCS is a robust indicator of injury severity and ciTBI.

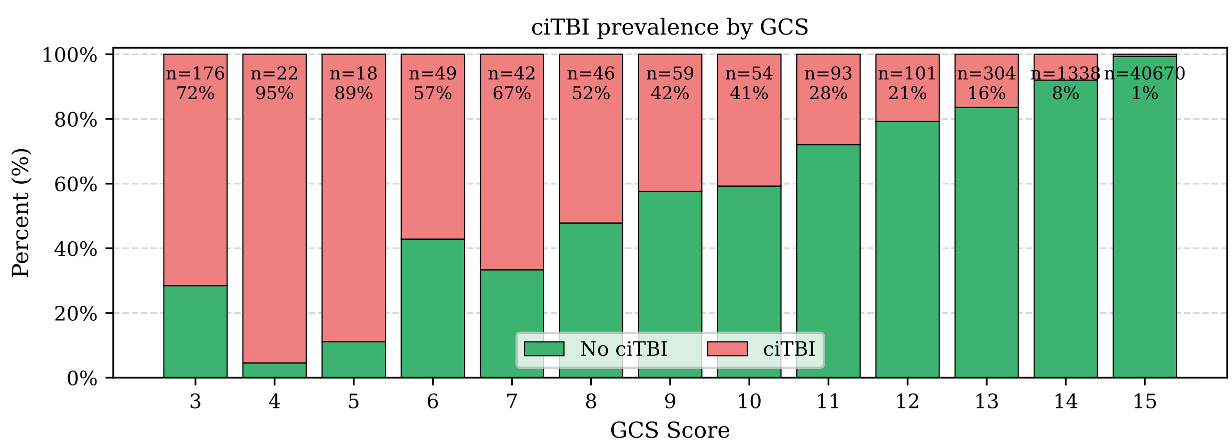


Figure 4: Relationship between Glasgow Coma Scale (GCS) and ciTBI across the full cohort. Age-stratified differences were minimal, so all patients are shown together. ciTBI prevalence is very low at GCS 15 and increases steadily with lower scores, highlighting GCS as a strong indicator of injury severity.

3 Findings

3.1 First finding

Figure 5 shows the correlations between boolean predictors and ciTBI, stratified by age group. Earlier, the grouped bar plots indicated which predictors tended to differ between ciTBI and non-ciTBI cases, while the categorical plots highlighted how missing values are concentrated in subjective variables. The correlation analysis provides a more quantitative summary, ranking predictors by their direct association with the outcome.

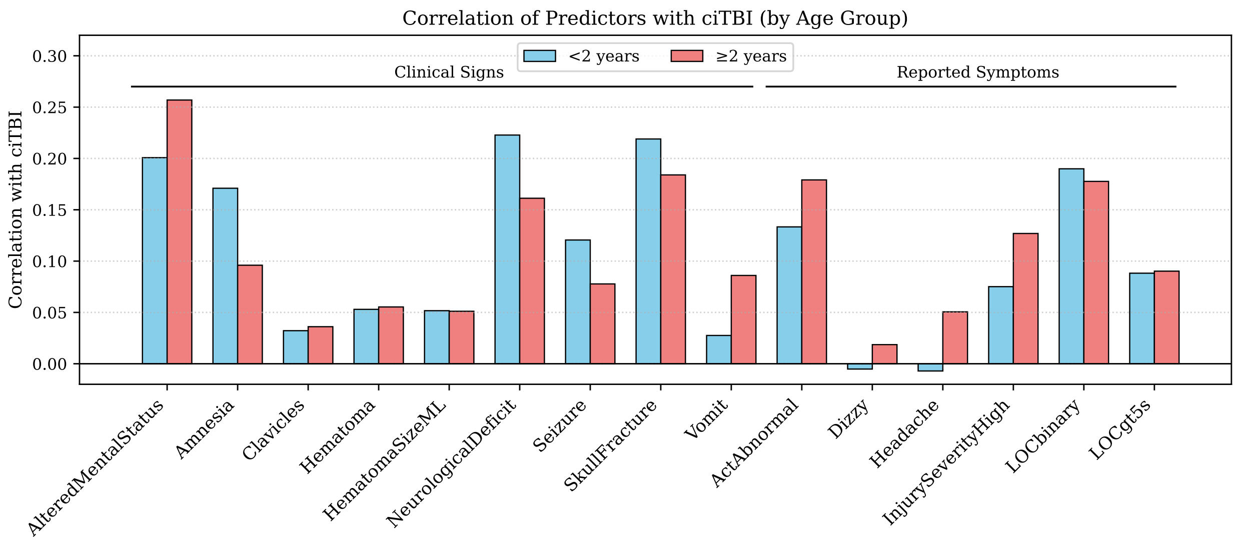


Figure 5: Correlations of boolean predictors with ciTBI. Strong predictors include `AlteredMentalStatus`, `NeurologicalDeficit`, `SkullFracture`, `Seizure`, and `ActAbnormal`, Loss of Conciousness (`LOCbinary`) with `Amnesia` additionally important in younger children. Reported symptoms such as Dizziness (`Dizzy`) or `Headache` as well as `Clavicles` and `Hematoma` show weak correlation.

Among the *clinical signs*, several variables stand out as consistently strong indicators across both age groups. `AlteredMentalStatus`, `NeurologicalDeficit`, and `SkullFracture` show the highest correlations with ciTBI, confirming their use in the PECARN rules [1]. `Seizure` also appears predictive, although the overall number of positive cases is low. By contrast, variables such as `Clavicles` or `HematomaSizeML` are weakly correlated, which aligns with Figure 3 and suggests limited standalone value.

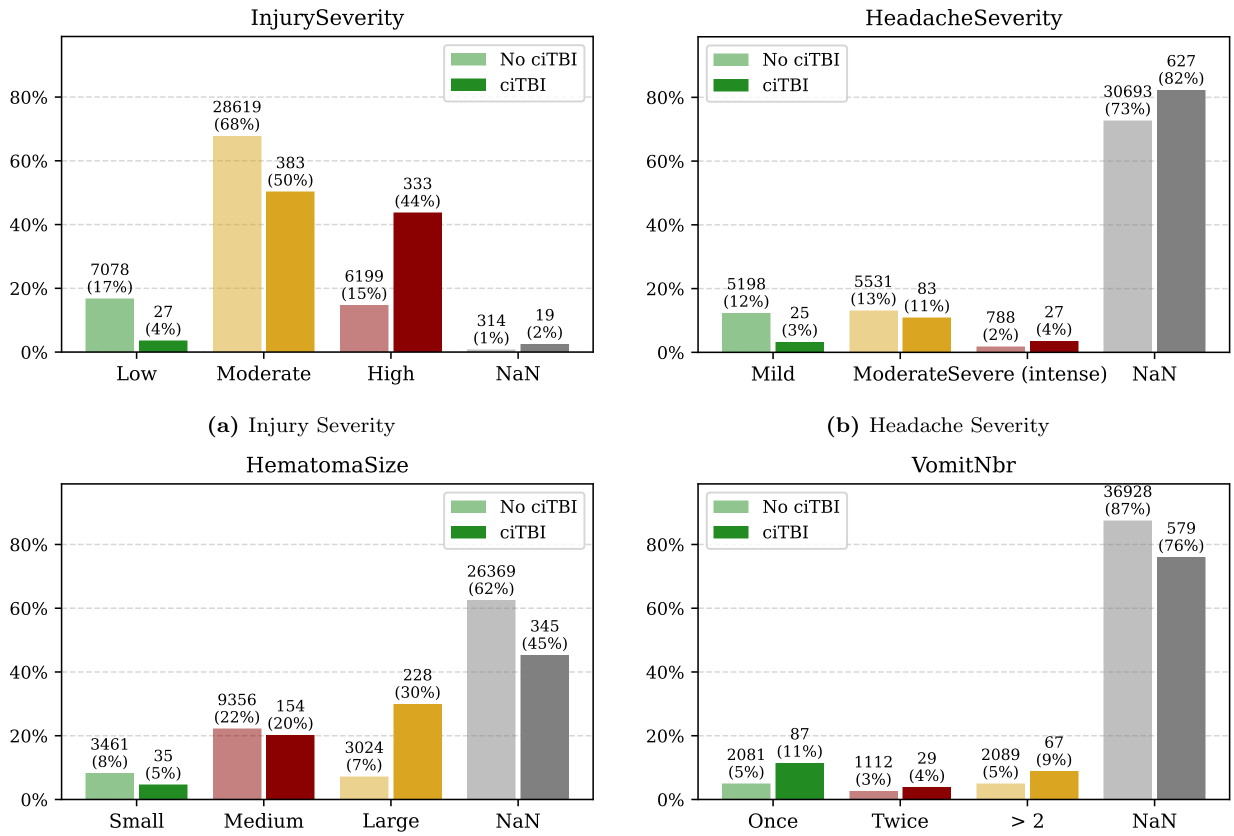
Within the *reported symptoms*, the overall correlations are weaker and more variable. `InjurySeverityHigh`, `LOCbinary`, `LOCgt5s`, and `ActAbnormal` show some association with ciTBI, while symptoms like `Headache` and `Dizzy` are minimally correlated. Importantly, these weaker variables are also those with high proportions of missing values, especially in the under-two group where subjective reporting is not feasible. This highlights the double challenge of such predictors: they are both less informative and less reliably recorded.

Taken together, these analyses suggest that a reduced predictor set focused on robust clinical signs is preferable. For both age groups, `AlteredMentalStatus`, `NeurologicalDeficit`, `SkullFracture`, and `Seizure` should be prioritized. In younger children, `Amnesia` is additionally informative, while in older children, reported variables such as `ActAbnormal` or `InjurySeverityHigh` can add value. Variables like `Headache` and `Dizzy`, given their weak associations and high missingness, appear less useful for modeling.

3.2 Second finding

Categorical predictors. Figure 6 summarizes the distributions of selected categorical predictors (`InjurySeverity`, `HeadacheSeverity`, `HematomaSize`, `VomitNbr`) and their relationship to ciTBI. Higher

`InjurySeverity` levels corresponds to higher ciTBI prevalence: 50% of moderate and 44% of high severity cases are ciTBIs, compared to just 4% of low severity cases. Similarly, both `HematomaSize` and `VomitNbr` suggest a dose–response pattern: larger hematomas and multiple vomiting episodes are associated with increased risk of ciTBI, even though the absolute number of ciTBI cases with the recorded symptom remains relatively small.



(a) Injury Severity

(b) Headache Severity

(c) Hematoma Size

(d) Vomit Number

Figure 6: Categorical predictors and their relationship to ciTBI. Higher levels of `InjurySeverity`, larger `HematomaSize`, and multiple `VomitNbr` episodes are associated with increased ciTBI prevalence, while `HeadacheSeverity` shows limited discriminatory power. Substantial proportions of missing values are present, especially for `HeadacheSeverity`

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`HeadacheSeverity`, on the other hand, provides only weak discrimination. Mild and moderate headache categories show very few ciTBI cases, and the overall number of reported severe headaches is small. Most importantly, more than 70% of entries for `HeadacheSeverity` are missing (NaN). This highlights a critical problem: missingness is not randomly distributed, but appears concentrated in variables that rely on patient self-report or parental observation. The same is true for `HematomaSize` and `VomitNbr`, where missing values represent around 70-90% of the sample. These high proportions of missing data limit the reliability of these predictors in isolation and can bias analyses if not handled appropriately.

The patterns of missingness themselves are therefore informative: the absence of data may reflect systematic biases, such as preverbal children being unable to report headache severity, or incomplete clinical documentation in certain contexts.

3.3 Reality Check

A useful way to validate the cleaned dataset is to compare it against published findings from the original PECARN study by Kuppermann et al. (2009). In that study, the prevalence of clinically important traumatic brain injury (ciTBI) was reported as approximately 0.9% across the 42,412 children included. Within the subgroup who got a CT scan, the ciTBI-positive cases were about 5%. These numbers provide an external benchmark for assessing whether the cleaned dataset is consistent with the real-world population.

For this reality check, the assumption is that the cleaned dataset is a representative subsample or extract of the full PECARN cohort. Therefore, overall proportions of outcomes and distributions of key predictors (e.g., GCS, altered mental status, injury severity) should be reasonably close to the reported values, but small deviations are expected due to missingness and the specific subset available.

When comparing the cleaned data to this reality, the overall prevalence of ciTBI in the dataset is 1.7%, which is broadly consistent with the 1% reported. Similarly, the rate of CT use and the proportion of positive CTs are of the same order of magnitude, suggesting that the dataset is capturing the same clinical patterns. However, some issues remain: Several variables, especially subjective symptoms such as headache severity and vomiting frequency, have very high proportions of missing values. This missingness likely reflects recording bias, as younger children cannot reliably self-report these symptoms. As a result, the dataset may underrepresent these predictors compared to the clinical reality.

3.4 Stability Check

The stability check focuses on how suspected loss of consciousness (LOC) values are coded. In the dataset, 1,997 observations (4.9% of non-missing LOC entries) were labeled as "suspected". When these were mapped as True versus mapped as False, the correlation between LOC and ciTBI changed only slightly in both age groups (Figure 7). This suggests that the treatment of Suspected LOC does not substantially alter the overall relationship between LOC and ciTBI, indicating that the finding is stable and robust to this specific recoding choice.

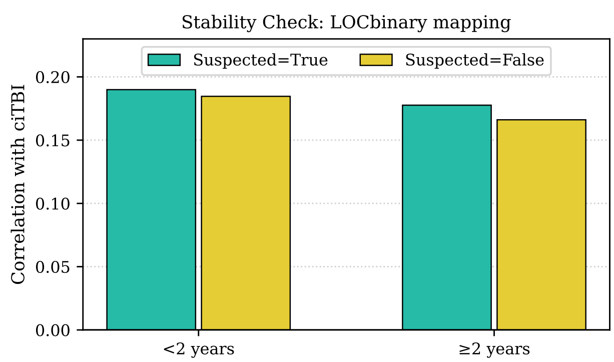

Figure 7: Stability check of LOC mapping. Correlation between LOC and ciTBI remains similar when Suspected LOC (4.9% of cases) is coded as True or False, indicating robustness of the finding across age groups.

4 Modeling

4.1 Implementation

Two supervised classification models were implemented to predict clinically important traumatic brain injury (ciTBI) and guide CT scan decisions. Logistic regression was selected as a baseline due to its interpretability and well-understood statistical properties. Random forest was chosen as a complementary model because of its flexibility in capturing non-linear interactions and its robustness to missingness and heterogeneous predictors.

In both models, preprocessing was handled through pipelines. Continuous variables such as age and GCS were imputed with medians and standardized, categorical variables were imputed with the most frequent value, and Boolean variables were imputed and cast to binary indicators. For logistic regression, the solver was set to lbfgs with class weights balanced to address class imbalance, and the maximum number of iterations was increased to 2000 to ensure convergence. For the random forest, 200 trees were grown with balanced class weights, and default depth was retained to allow the ensemble to determine complexity. Hyperparameters were chosen to balance reproducibility with computational efficiency; no extensive grid search was performed, but values were guided by standard practice in the literature.

4.2 Stability

To evaluate stability, the perturbation from Section 3.4 was applied by re-coding the treatment of suspected loss of consciousness (LOC). Both models were then retrained and evaluated on the perturbed data. The ROC curves in Figure 8 show that predictive behavior remained unchanged: logistic regression achieved an AUC of 0.93 on both the original and perturbed datasets, and the random forest achieved an AUC of 0.90 in both cases. The near-perfect overlap of the curves indicates that the models are robust to this reasonable preprocessing variation.

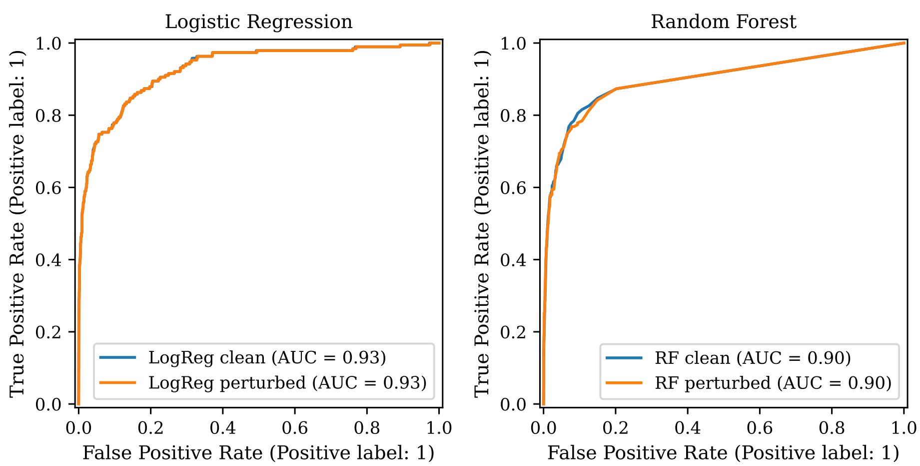

Figure 8: Stability check ROC curves for logistic regression (left) and random forest (right). Each panel shows the ROC curve for the model trained on the original cleaned data (blue) and on the perturbed data (orange), where the definition of suspected LOC was altered. Both models achieved identical AUC values (0.93 and 0.90 respectively) under the perturbation, indicating that prediction is stable to this specific change in the preprocessing.

4.3 Discussion

Logistic regression was chosen because it provides an interpretable, parametric summary of how clinical variables affect ciTBI risk, consistent with the PECARN approach of simple decision rules. Its coefficients can be directly inspected to assess whether predictors such as altered mental status or skull fracture increase estimated risk. Random forest was added to capture complex interactions that may be missed by a linear model, thereby serving as a robustness check.

While logistic regression is straightforward to interpret, the random forest is not inherently transparent. Its predictions can, however, be explained through feature importance measures and visualization of partial dependence plots. Clinicians may find the logistic regression more immediately useful for decision support, but the random forest highlights potential non-linear effects worth further investigation. Together, the two models provide complementary insights: one prioritizing interpretability, the other flexibility.

5 Conclusion

In this lab, the PECARN dataset was used to investigate predictors of clinically important traumatic brain injury (ciTBI) in children and to reflect on the PCS framework. The data/reality stage revealed substantial cleaning requirements: many variables needed to be recoded, implausible entries were removed, and derived indicators such as LOCbinary, InjurySeverityHigh, and HematomaSizeML were constructed. Exploratory analysis showed that key predictors such as altered mental status, skull fracture, and severe mechanisms of injury were strongly enriched among ciTBI cases, while other symptoms such as vomiting and headache were common but less useful.

From a modeling perspective, both logistic regression and random forest classifiers were evaluated. Logistic regression, consistent with PECARN's original decision rules, achieved strong discrimination with an AUC around 0.93, while the random forest performed similarly with an AUC around 0.90. The ROC curves confirmed that both models could identify high-risk children effectively, and stability checks (e.g., perturbing how loss of consciousness was coded) showed that predictive performance was robust to reasonable data processing choices.

As an outlook, the outcome of the analysis highlights both progress and limitations. Comparing unnecessary vs. necessary CT scans showed that a large fraction of children underwent CT despite not having ciTBI, while only a small proportion of scans revealed clinically important injuries. This imbalance reflects the fundamental clinical dilemma: avoiding missed cases while limiting unnecessary radiation. The findings reinforce the validity of the established PECARN rules while also suggesting that more flexible models could marginally improve prediction. At the same time, variables with high missingness or subjective measurement (e.g., headache severity) limited model usefulness, pointing to the need for better future data collection.

Overall, this lab demonstrated that careful data cleaning, visualization, and modeling can reproduce known results and provide additional insights. The PCS framework ensured that issues of reality (data quality), predictability (model performance), and stability (robustness to perturbations) were addressed systematically. The main outcome is a reaffirmation that simple, interpretable predictors remain highly effective for identifying children at risk of ciTBI, while highlighting opportunities for reducing unnecessary CT scans.

6 Academic honesty statement

I affirm that this report represents my own work. All sources of information, data, figures, and code that are not my own have been properly cited and acknowledged. No unauthorized assistance was received in the completion of this assignment, and I have abided by the principles of academic integrity as required by the course and the institution.

7 Bibliography

References

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